Answer Key - Mathematics - Undergraduate Level Assessment 4 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** B
Options: A) -5; B) -4; C) -3; D) -2
Note: The correct answer is -4 because the equation 7 m + 12 n = 22 can be rewritten to express m in terms of n, allowing us to find integer pairs (m, n) that satisfy the equation, and among the possible sums m + n, the maximum negative sum occurs when m = -4 and n = 0, resulting in m + n = -4.

2. **Answer:** D
Options: A) None; B) One; C) Four; D) Five
Note: The number of trailing zeros in k! is determined by the formula k/5 + k/25 + k/125, and for k! to end in exactly 99 zeros, this formula yields five specific values of k, specifically for k = 400, 401, 402, 403, and 404.

3. **Answer:** B
Options: A) 8; B) 120; C) 240; D) 24
Note: The maximum possible order of an element in the direct product of groups is the least common multiple of the orders of the individual groups, so for Z_8 (order 8), Z_10 (order 10), and Z_24 (order 24), the LCM is 120.

4. **Answer:** B
Options: A) 1; B) 2; C) 3; D) 4
Note: The number 2 is a generator for the finite field Z_11 because it generates all non-zero elements of the field when raised to the powers of integers from 0 to 9, demonstrating that it is a primitive root modulo 11.

5. **Answer:** D
Options: A) (S, +, x) is closed under addition modulo 10.; B) (S, +, x) is closed under multiplication modulo 10.; C) (S, +, x) has an identity under addition modulo 10.; D) (S, +, x) has no identity under multiplication modulo 10.
Note: The statement is correct because the subset S = {0, 2, 4, 6, 8} lacks a multiplicative identity element under multiplication modulo 10, as none of its elements can multiply with every other element in S to yield an element of S itself that serves as the multiplicative identity (which would be 1 in a ring).

True / False

6. **Answer:** True
Options: A) True; B) False
Note: The statement is true because the three nonisomorphic trees with 5 vertices are characterized by their distinct structures: one with a star configuration (1 central

vertex connected to 4 leaves), one with a linear configuration (a path of 5 vertices), and one with a more branched configuration (a vertex connected to two others, which each have one additional leaf).

7. **Answer:** False
   Options: A) True; B) False
   Note: The statement is false because the subgroup 4 Z has multiple cosets in 2 Z, specifically 4 Z and 2 Z 4 Z, which includes elements like 2 and 6.

8. **Answer:** True
   Options: A) True; B) False
   Note: The product of 20 and -8 in the ring Z_26 is calculated as (20 * -8) mod 26, which equals -160 mod 26, resulting in 22.

9. **Answer:** False
   Options: A) True; B) False
   Note: The largest order of an element in the symmetric group S_5, which represents the permutations of 5 objects, is actually 60, corresponding to the permutation with the cycle structure of a 5-cycle.

10. **Answer:** False
    Options: A) True; B) False
    Note: The statement is false because the correct probability that 0 will not be chosen when selecting k digits is $(9/10)^k$, as there are 9 other digits (1 through 9) that can be chosen for each selection.

Long Form Response

11. **Answer:** Given a real 2 x 2 matrix A such that A = $A^{(}$-1) and I $\neq$ A $\neq$ -I, we can derive implications for the trace of A. Since A is its own inverse, it must satisfy A^2 = I, meaning the eigenvalues of A are either 1 or -1. However, since A is neither the identity matrix (I) nor the negative identity matrix (-I), it must have one eigenvalue of 1 and one of -1. The trace of a matrix is the sum of its eigenvalues, so in this case, the trace of A is 1 + (-1) = 0. Thus, our analysis reveals that the trace of matrix A is 0.
    Note: correct because it logically deduces that since the matrix A is its own inverse, its eigenvalues must be 1 and -1, leading to a trace of 0, while also adhering to the given conditions that A is neither the identity nor the negative identity matrix.

12. **Answer:** To determine the minimum value of the expression x + 4 z under the constraint $x^2 + y^2 + z^2 \leq 2$, we can utilize the method of Lagrange multipliers or analyze the geometry of the constraint. Since the constraint describes a sphere of radius $\sqrt{}$ 2, we can parametrically express x and z in terms of spherical coordinates. By substituting z = (2 - x^2 - y^2)$^{(}$1/2) into the expression and optimizing, we find that the minimum occurs when x and z take on specific values. After calculations, we find that the minimum value of the expression x + 4 z is -$\sqrt{}$ 34, which is achieved when x =

-1 and z = -$\sqrt{2}$. This solution illustrates the relationship between optimization and geometric constraints in multivariable calculus.
Note: correct because it employs the method of Lagrange multipliers or geometric analysis to optimize the expression x + 4 z within the given spherical constraint, leading to specific values for x and z that yield the minimum.

End of Answer Key